were prepared to envisage under the old, departing paradigm of adaptation for the benefit of the species. It is this subtlety, roughly, that is the theme of the rest of the book. Take a simple example, birdsong. It seems a very inefficient arrangement: a naïve materialist looking for the techniques by which a species of *Turdus* survives hard winters, food shortages, and the like might well find the flamboyant singing of its males as improbable as ectoplasm at a séance. (On further thought he might find the fact that the species has males at all equally improbable, and this indeed is another major topic of the book: as with that of birdsong, the function of sex has been rationalized much too facilely in the past.) Yet within any bird species a whole team of replicators has concerned itself to lay down an elaborate outline for this performance. Somewhere Dawkins cites the even more extraordinary song of the humpback whale, which may make itself heard over a whole ocean; but of this song we know even less than with *Turdus* what it is about and to whom directed. So far as the evidence goes it might actually be an anthem for cetacean unity against mankind—perhaps well for whales if it were. Of course, it is other teams of teams of replicators that now turn out symphony concerts. And these certainly do sometimes cross oceans —by reflection from bodies in space which themselves were made and orbited according to plans from even more complex teams. What conjurers do with mirrors is nothing to what nature, if Dawkins is right, does with no more promising a starting material than congealed primeval soup. It will serve to characterize the new look that biology has in this and some other recent books (such as E. O. Wilson's *Sociobiology*) to say that it shines with a hope that these farthest extensions of life may soon fit more comprehensibly, in essence if not in some details (religious persons and Neo-Marxists may reverse that phrase if it suits them better), into a general pattern that includes the simplest cell wall, the simplest multicell body, and the blackbird's song.

The impression should be avoided, however, that this book is some sort of layman's or poor man's *Sociobiology*. First, it has many original ideas, and second, it counterweights a certain imbalance in Wilson's massive tome by strongly emphasizing the game-theoretic aspect of social behavior, which Wilson hardly mentioned. 'Game-theoretic' is not quite the right word, especially in the context of lower levels of social evolution, since the genes themselves don't rationalize about their methods of operation; nevertheless, it has become clear that at all levels there are useful similarities between the conceptual structures of game theory and those of social evolution. The cross-fertilization implied here is new and is still in progress: only recently, for example, I learned that game theory had already given a name ("Nash equilibrium") to a concept that corresponds roughly to the "evolutionarily stable strategy." Dawkins rightly treats the idea of evolutionary stability as all-important for his new overview of social biology. The game-like element in social behavior and social adaptation comes from the dependence, in any social situation, of the success of one individual's strategy on the strategies used by his or her interactants. The pursuit of adaptation that gets the most out of a given situation regardless of the overall good can lead to some very surprising results. Who would have supposed, for example, that the weighty matter of why in fish, contrary to the case in most other animals, it is the male that usually guards the eggs and young if either sex does, might depend on such a trivial detail as which sex is constrained to release its gametes into the water first? Yet Dawkins and a co-worker, pursuing an idea of R. L. Trivers's, have made a fair case that such a detail of timing, even if a matter of seconds, could be crucial for the whole phenomenon. Again, would we not expect that females of monogamous birds, blessed with the help of a mate, would lay larger clutches than females of polygamous species? Actually the reverse is true. Dawkins, in his somewhat alarming chapter on the "battle of the sexes," applies once more the idea of stability against exploitation (by the male in this case) and suddenly makes this odd correlation seem natural. His idea, like most of his others, remains unproven, and there may well be other, more weighty reasons; but the ones he gives, which are seen so easily from his new vantage point, demand notice.

In a textbook of game theory one sees no more of games than one sees of circles and triangles in a textbook of modern geometry. At a glance all is just algebra: game theory is a technical subject from the start. Thus it is certainly a literary feat to convey as much as this book does of